

Arnav Ajay

Project Portfolio: <https://arnav-04-ajay.github.io> | LinkedIn: @arnav-ajay04 | arnav.ajay@duke.edu

EDUCATION

Duke University

Aug. 2022 – May 2026

BS in Computer Science and BSE in Electrical and Computer Engineering, GPA: 3.85

Durham, NC

EXPERIENCE

Electrical Engineering and Embedded Systems Intern

May 2025 – August 2025

Stryker Instruments R&D

Portage, MI

- Created and tested new vacuum regulator system for upcoming iterations of flagship product Neptune®
- Spec'd new motors and developed and implemented custom motor drive circuitry and PID control (STM32 based) aimed at higher long term durability and power efficiency
- System I developed passed testing plan and exceeded expectations when integrated into the existing Neptune® system. Shows strong potential for implementation in future versions

Robotics Engineering Intern

May 2024 – August 2024

Stryker Surgical Robotics R&D

Weston, FL

- Developed relative positioning system for new surgical robotics application. System performed point pair transformations with emphasis on accuracy and repeatability
- Utilized SolidWorks to design and simulate custom parts for surgical-specific relative positioning
- Leveraged MATLAB and Python to analyze position data, and conduct a Gage R&R analysis of the positioning system. Translational and rotational errors were under 2mm and 2 degrees respectively (within surgical spec)
- System passed cadaver testing and the full surgical system has received positive feedback during limited release

iOS App Development Intern

June 2023 – July 2024

Fyzical Therapy & Balance

Durham, NC

- Working under Professor Rahib Younes to develop and deliver 3 unique iOS physical therapy apps
- Designed mobile exercises with Swift and implemented Apple Vision and ARKit libraries in SwiftUI to create physical therapy exercises with live pupil and gaze tracking features to assist with Oculomotor Disorder

Electrical Engineering Contractor

Sep. 2022 – Aug. 2023

Durham Museum of Life and Science

Durham, NC

- Collaboratively designed a custom humidification system for the museum's Leafcutter Ant exhibit. Goal of sustained 80% humidity was achieved
- Implemented a PID control of the exhibit's humidity via ESP32, and hosted a web server for humidifier status
- Designed 3D printed housings in SolidWorks, and used Altium to produce a custom PCB for the product

EXTRACURRICULARS

Duke Aero Payload EE Team Lead | ESP32 Pro S3, STM32, Altium, KiCad, C

Aug. 2024 – Present

- Tasked with design of fully custom flight computers for the rocket payload, which oversees live telemetry transmission, power distribution, and communication across all subsystems
- Developing a system to capture live digital video for onboard storage as well as live transmission
- Custom controls systems are being put in place for guided recovery and wind-based energy recycling

Digital Systems / FPGA Teaching Assistant | Artix A-7, Verilog, Processor Design

January 2025 – Present

- Guide students as they work through designing a 5-stage pipelined processor in Verilog and testing on an FPGA
- Mentor two month-long final projects per semester that utilize the finished processor and interface with custom I/O (custom hardware drivers written in Verilog) to achieve real-time processing tasks

Multimodal Machine Learning Research | TensorFlow, Keras, PyTorch, VQA

Sept 2021 – May 2022

- Worked under Professor Farshad Firouzi to build a Multimodal model for chest X-ray VQA tasks
- Constructed a multimodal learning framework using a combined vision-language encoder with self-supervised pre-training for this task (training accuracy over 90%)

SKILLS

Software: Flask, REST APIs, PyTorch, TensorFlow, Xcode, Docker, Java, Python, C/C++, Swift, MATLAB, Verilog
Electrical Engineering Tools: Embedded Microcontrollers(ESP32, Arduino, Raspberry Pi, STM32), PCB Design (Altium, KiCad), PSpice, MATLAB, Simulink, CAD (AutoCAD, Fusion360, SolidWorks), Product Design